Assignment 1 (2024111017)

This report is not AI generated, only code is (as per rules) https://github.com/orbi8r/LMAO-A1

Document: Catan game rulebook (16 pages, 86 chunks of 120 words) idk why I had this pdf lying around.

Models: all-MiniLM-L6-v2 (bi encoder), ms-marco-MiniLM-L-6-v2 (cross encoder), Qwen2.5-1.5B-Instruct (Huge LLM ik, same for both pipelines). everything ran on a cloud CPU node.

1. Architecture comparison

A bi encoder embeds the query and each chunk separately. Chunk vectors are computed Offline stage and relevance is the cosine similarity between two fixed vectors. a cross encoder reads the query and one chunk together as a single input, so self attention can compare every query, chunk token. it gives a more accurate relevance score, but it needs a full online step for every (query, chunk) pair.

2. Qualitative and rank shift analysis

Read the ipynb

```
queries = [
 [
  "Exception",
  "If two players tie for longest road, who holds the card?",
  "two or more players tie for the new longest road"
 ],
 [
  "Buried chunk",
  "When can victory point cards be shown?",
  "You only reveal victory point cards when the game is over"
 ],
 [
  "Keyword trap",
  "Who goes first in the game?",
  "the oldest player goes first"
 ],
 [
  "Negation",
  "Is it forbidden to reveal victory point cards early?",
  "You only reveal victory point cards when the game is over"
 ],
 [
  "Buried chunk",
  "Who can steal a card when the robber is moved?",
  "you also steal 1 resource card at random"
 ],
 [
  "Exception, both fail",
  "Can I play the development card I just bought?",
  "may not be a card you bought during the same turn"
 ],
 [
  "Control",
  "Do I lose cards when someone else rolls a seven and I only have six?",
  "has only 6 cards in his hand"
 ]
]
```

search ranks (base -> reranked):

- longest road tie: 5 -> 1
- show VP cards: 9 -> 2
- who start game: 5 -> 1
- forbidden show VP early: 4 -> 3
- robber steal: 4 -> 2
- play dev card bought same turn: fail, not even in top 10 either way so rerank cant help

- control check (lose cards on 7): rank 1 both

cross encoder pull 5 queries from outside top 3 straight into top 3. dev card only one totally miss top 10.

	label	query	key_chunk	rank_baseline	rank_rerank
0	Exception	If two players tie for longest road, who holds...	39	5.0	1.0
1	Buried chunk	When can victory point cards be shown?	28	9.0	2.0
2	Keyword trap	Who goes first in the game?	9	5.0	1.0
3	Negation	Is it forbidden to reveal victory point cards ...	28	4.0	3.0
4	Buried chunk	Who can steal a card when the robber is moved?	52	4.0	2.0
5	Exception, both fail	Can I play the development card I just bought?	18	NaN	NaN
6	Control	Do I lose cards when someone else rolls a seve...	51	1.0	1.0

answers (base A1 vs rerank A2):

- road tie: A1 make up fake rule (1 pt each).A2 find right line (card put aside) but misread it, say both hold card; better proof, answer still wrong.
- show VP: A1 no key chunk,invent "exact 10 pts end of turn". A2 get main idea right (show when sure 10pts) but add wrong extra stuff.
- who go first: both say the highest player goes first. and miss real rule (oldest player first).
- forbidden reveal early: both say yes. A1 make up sentence not in book. A2 add correct exception (hitting 10pts).
- robber steal: A1 actually right. A2 totally garbled. rerank make it worse here.
- dev card & control: both wrong across board. funniest part is control chunk literally at rank 1 and 1.5B model still misread it lol.

reranker pulled the key chunk into top 3 for 5 of 7 queries (dev card missed top 10 for both, control was already rank1), but 1.5B model just too tiny and dumb to read right even when chunk right in front of it. need bigger model to actually see gains. but fairs, just wanted to test.

```
[Exception] If two players tie for longest road, who holds the card?
  key chunk 39: baseline rank 5 → reranked rank 1
  Answer 1: If two players tie for the longest road, neither player will hold the "Longest Road" card because each player would need to have built a road that was exactly as long as the other's longest road to be eligible for the card. The rules state that if there is a tie for the longest road, no one wins the card; instead, both players receive a 1 victory point each.
  Answer 2: If two players tie for the longest road, both players would hold the "Longest Road" card. According to the rulebook excerpt provided:  "If you no longer have the longest road, but two or more players tie for the new longest road, set the 'Longest Road' card aside."  This means that once there's a tie for the longest road, the cards remain with those players until someone breaks the tie.
[Buried chunk] When can victory point cards be shown?
  key chunk 28: baseline rank 9 → reranked rank 2
  Answer 1: Victory point cards can be revealed when you or someone else wins the game, or if you have exactly 10 victory points at the end of your turn.
  Answer 2: Victory point cards can be revealed when either:  1. During your own turn, specifically when you reach exactly 10 victory points. 2. After revealing them yourself, if you're confident you have 10 or more victory points. 3. When another player wins the game, even if they don't reach 10 points yet.
[Keyword trap] Who goes first in the game?
  key chunk 9: baseline rank 5 → reranked rank 1
  Answer 1: The highest-ranking player starts the game as the starting player and begins.
  Answer 2: The highest player goes first in the game.
[Negation] Is it forbidden to reveal victory point cards early?
  key chunk 28: baseline rank 4 → reranked rank 3
  Answer 1: Yes, according to the rulebook excerpt provided, victory point cards must be kept hidden until the end of the game unless explicitly stated otherwise. The rule states "Keep victory point cards hidden until the end of the game." Therefore, revealing victory point cards before this time would be against the rules.
  Answer 2: Yes, according to the rulebook excerpt provided, victory point cards must be kept hidden until the end of the game unless revealed due to reaching exactly 10 victory points or being declared as the winner. Therefore, it is not allowed to reveal victory point cards early under normal circumstances.
[Buried chunk] Who can steal a card when the robber is moved?
  key chunk 52: baseline rank 4 → reranked rank 2
  Answer 1: When the robber is moved, any player who has a settlement or city adjacent to the target terrain hex can steal a card from the player who was robbed. However, if the target hex is adjacent to two or more players' settlements or cities, the player chooses which one to steal from.
  Answer 2: When the robber is moved, a player who has a settlement or city adjacent to the target terrain hex receives a resource card stolen from them.
[Exception, both fail] Can I play the development card I just bought?
  key chunk 18: baseline rank None → reranked rank None
  Answer 1: Yes, you can play the development card you just bought as long as it meets the criteria for playing a development card according to the rules provided. Specifically:  - You can play 1 development card per turn. - The card must be played after rolling the dice. - You cannot play a card that was bought during the same turn. - A victory point card Y that brings you to a certain number of points can also be used as a development card under specific conditions.  So, once you have met all these requirements, you can indeed play the card you purchased.
  Answer 2: Yes, you can play the development card you just bought as long as it meets the conditions for playing a development card according to the rules provided. Specifically:  - You can play the card at any time, including before rolling the dice. - However, you cannot play a card that was bought during the same turn. - Additionally, if you buy a card and it's a victory point card Y that brings you to a certain condition (like reaching 7 resource cards), you need to follow those specific rules related to building with those resources.
[Control] Do I lose cards when someone else rolls a seven and I only have six?
  key chunk 51: baseline rank 1 → reranked rank 1
  Answer 1: Yes, according to the rulebook excerpt provided, if you roll a "7" and have fewer than 7 resource cards, you will need to discard half of your remaining cards. In this case, since you only have 6 cards, you would discard 3 cards (half of 6), returning the discarded cards back into the supply stack.
  Answer 2: Yes, according to the rulebook excerpt provided, if you roll a "7" and have fewer than 7 resource cards, you will need to discard half of those cards. In this case, since you only have 6 cards, you would discard 3 cards (half of 6 rounded down), returning the remaining 3 cards to the supply stack.
```

3. Latency profiling

speed stats too (avg 7 queries, no llm time):

baseline bi encoder: 0.0081s

rerank (search + cross encoder top 10): 0.0689s

reranking like 8times slower per query. full per query numbers sitting in last table of notebook if u wanna check.

label	baseline (bi encoder search)	re ranking pipeline (search + cross encoder on 10 pairs)
Exception	0.0062	0.0708
Buried chunk	0.0106	0.0741
Keyword trap	0.0061	0.0598
Negation	0.0105	0.0763
Buried chunk	0.0064	0.0617
Exception, both fail	0.0094	0.0750
Control	0.0077	0.0646
mean	0.0081	0.0689

4. Why not run the cross encoder over the whole collection

cross encoder cant precompute chunk scores ahead of time. score needs query+chunk together, so every query needs a full pass per chunk. scales linear with doc count. 86 chunks is fine but millions of chunks? lol rip way too slow

bi encoder way faster; embeds whole collection once upfront.when query comes, just 1 embedding + instant vector search matmul.

so trick is: two stage, bring it to K candidates then cross encoder.